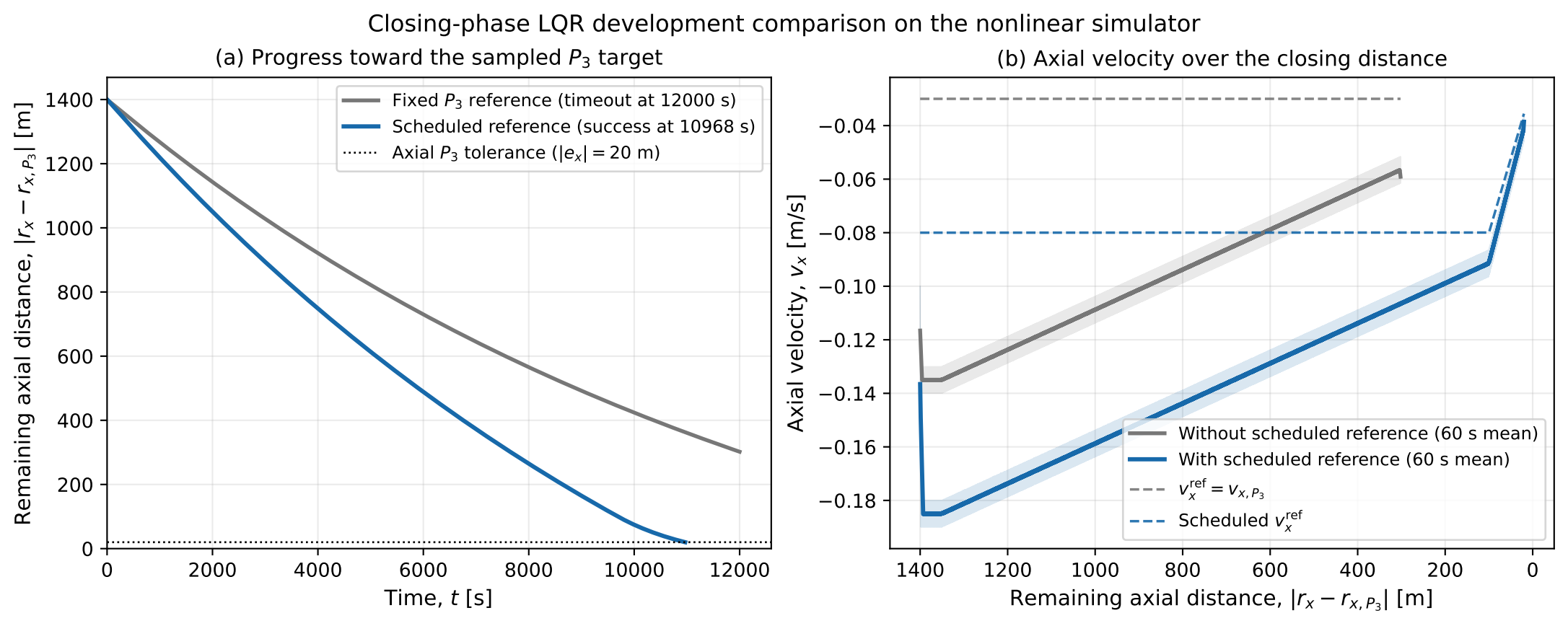

Closing-phase LQR development comparison on the nonlinear simulator
(a) Progress toward the sampled P_3 target
Fixed P_3 reference (timeout at 12000 s)
Scheduled reference (success at 10968 s)
Axial P_3 tolerance ($|e_x| = 20$ m)
Remaining axial distance, $|r_x - r_{x,P_3}|$ [m]
Time, t [s]
(b) Axial velocity over the closing distance
Without scheduled reference (60 s mean)
With scheduled reference (60 s mean)
$v_x^{\mathrm{ref}} = v_{x,P_3}$
Scheduled v_x^{ref}
Axial velocity, v_x [m/s]
Remaining axial distance, $|r_x - r_{x,P_3}|$ [m]